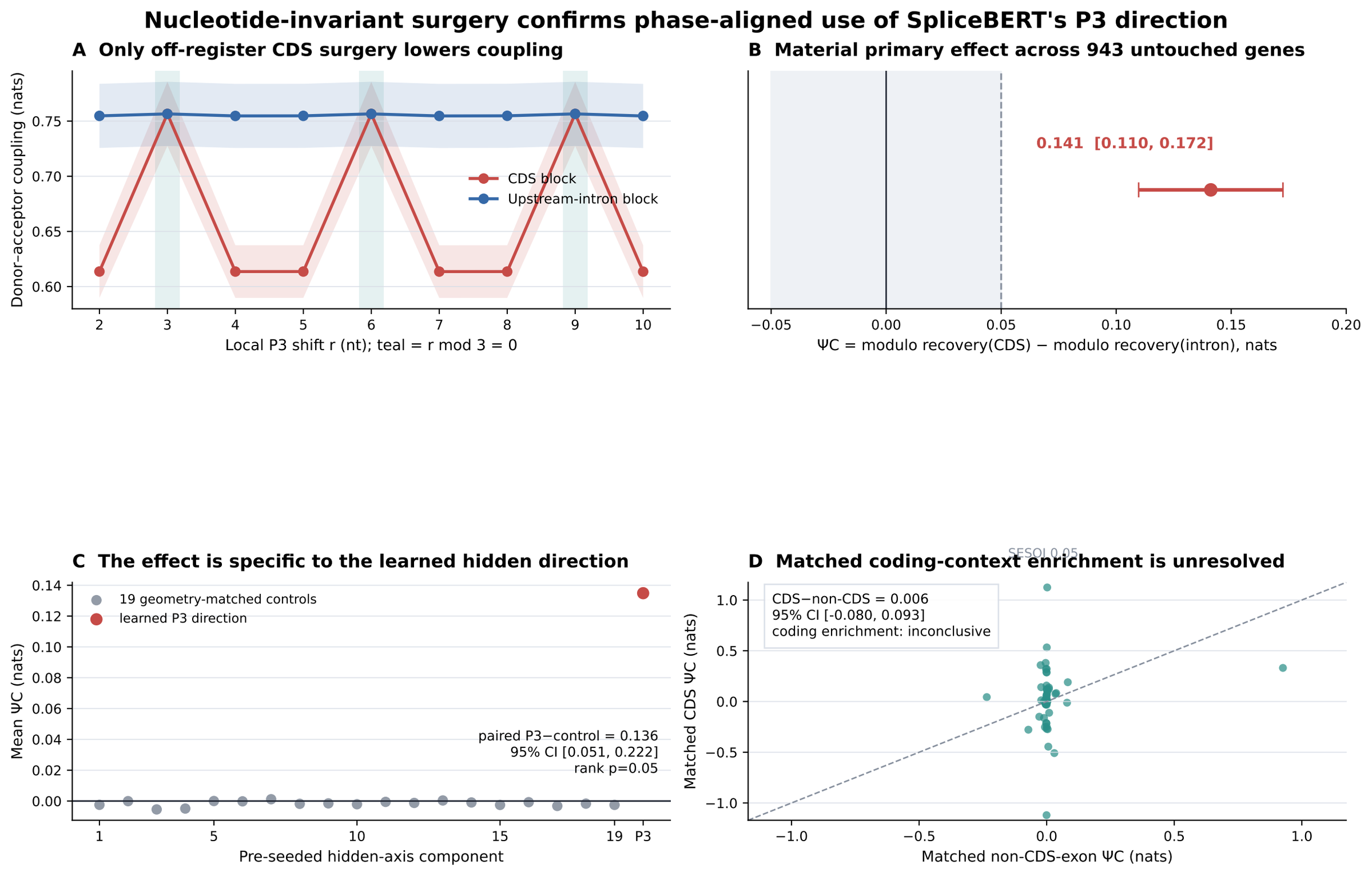
Nucleotide-invariant surgery confirms phase-aligned use of SpliceBERT's P3 direction
A Only off-register CDS surgery lowers coupling
Donor–acceptor coupling (nats)
0.75
0.70
0.65
0.60
CDS block
Upstream-intron block
2
3
4
5
6
7
8
9
10
Local P3 shift r (nt); teal = r mod 3 = 0
B Material primary effect across 943 untouched genes
0.141 [0.110, 0.172]
−0.05
0.00
0.05
0.10
0.15
0.20
ΨC = modulo recovery(CDS) − modulo recovery(intron), nats
C The effect is specific to the learned hidden direction
Mean ΨC (nats)
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
19 geometry-matched controls
learned P3 direction
paired P3−control = 0.136
95% CI [0.051, 0.222]
rank p=0.05
1
5
10
15
19
P3
Pre-seeded hidden-axis component
D Matched coding-context enrichment is unresolved
SESOI 0.05
CDS−non-CDS = 0.006
95% CI [-0.080, 0.093]
coding enrichment: inconclusive
Matched CDS ΨC (nats)
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
Matched non-CDS-exon ΨC (nats)